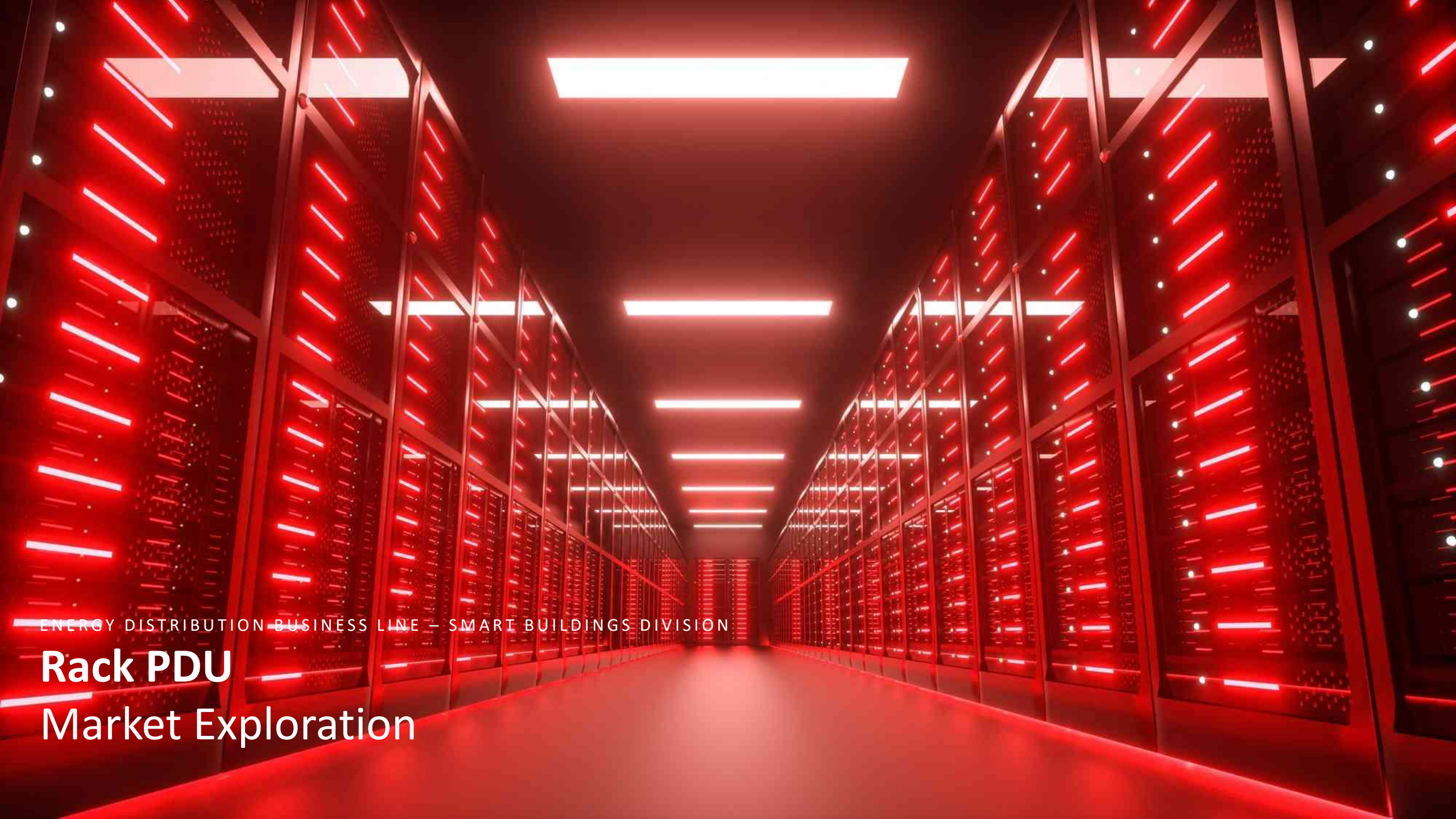




ENERGY DISTRIBUTION BUSINESS LINE – SMART BUILDINGS DIVISION

Rack PDU

Market Exploration

ABB Rack PDU Solution

Power distribution levels in data centers

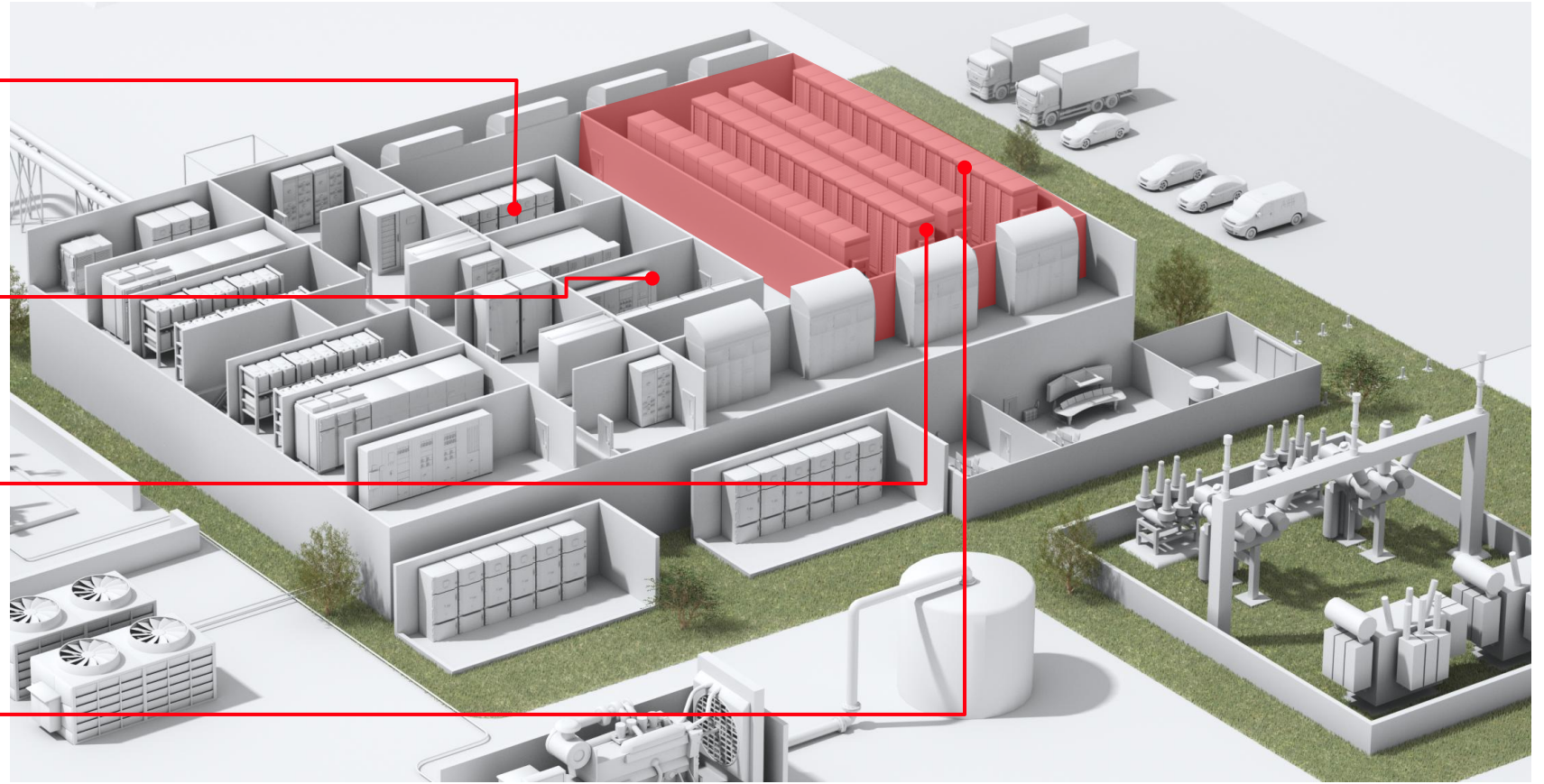
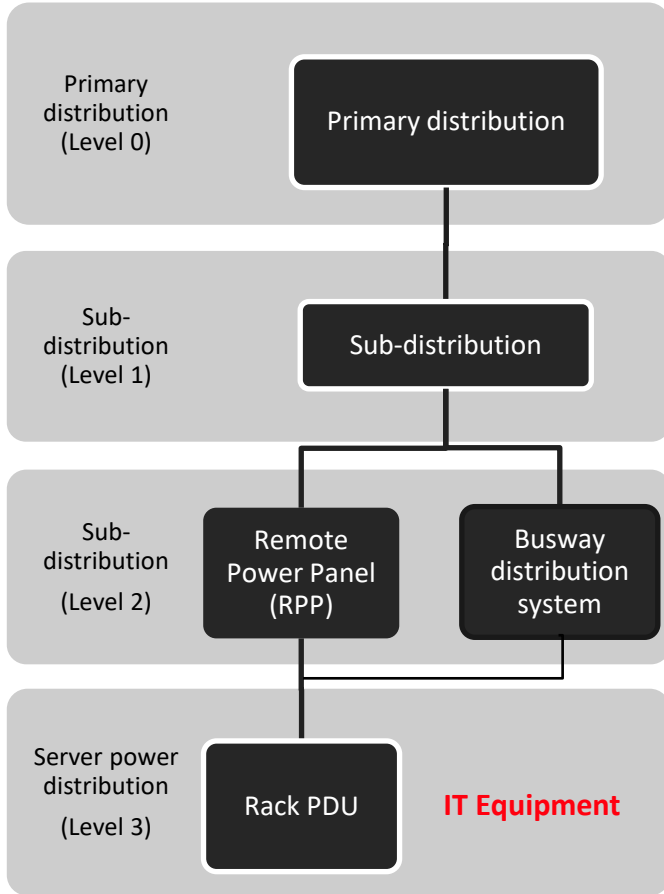
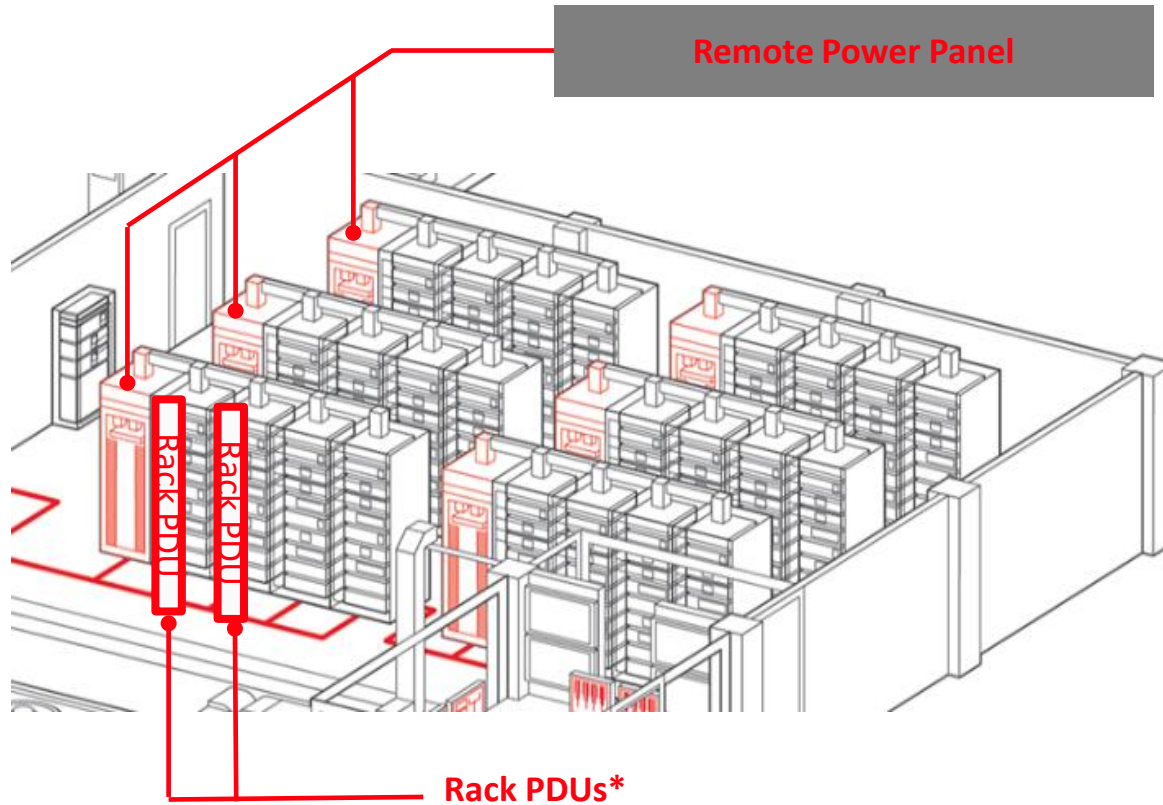


ABB Rack PDU Solution

Remote Power Panel vs. Busbar+Tapoff units



*or power shelves

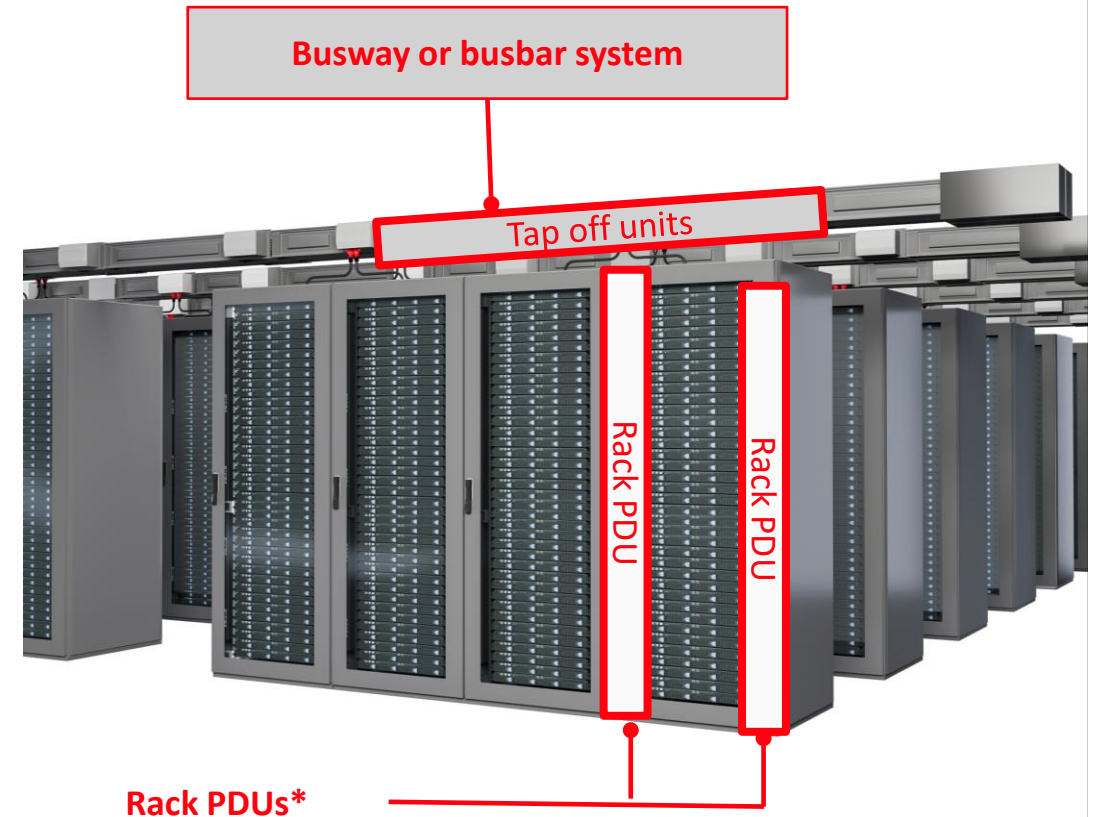
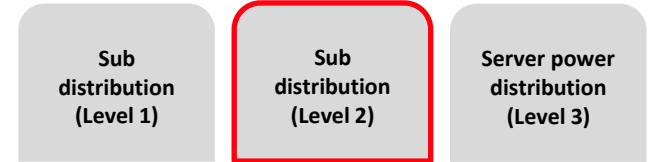


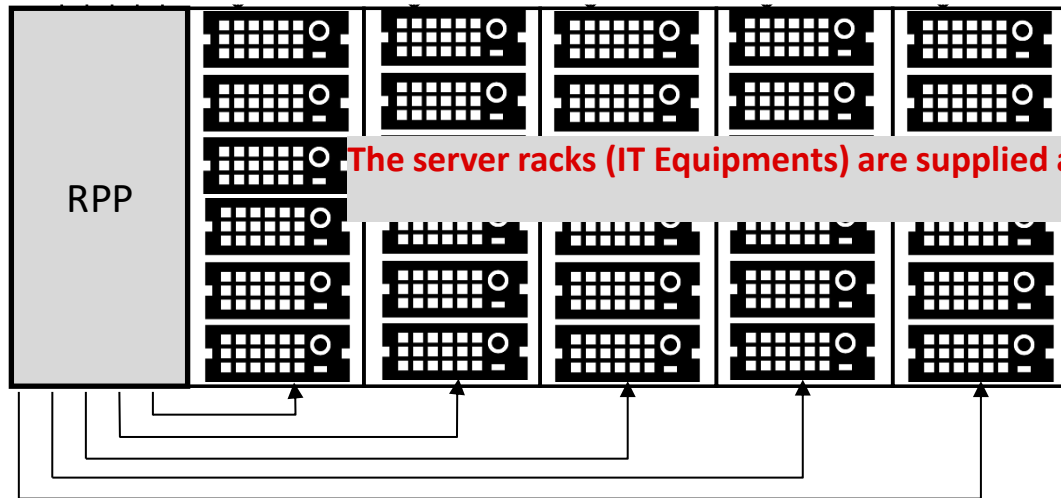
ABB Rack PDU Solution

Remote Power Panel vs. Busbar+Tapoff units



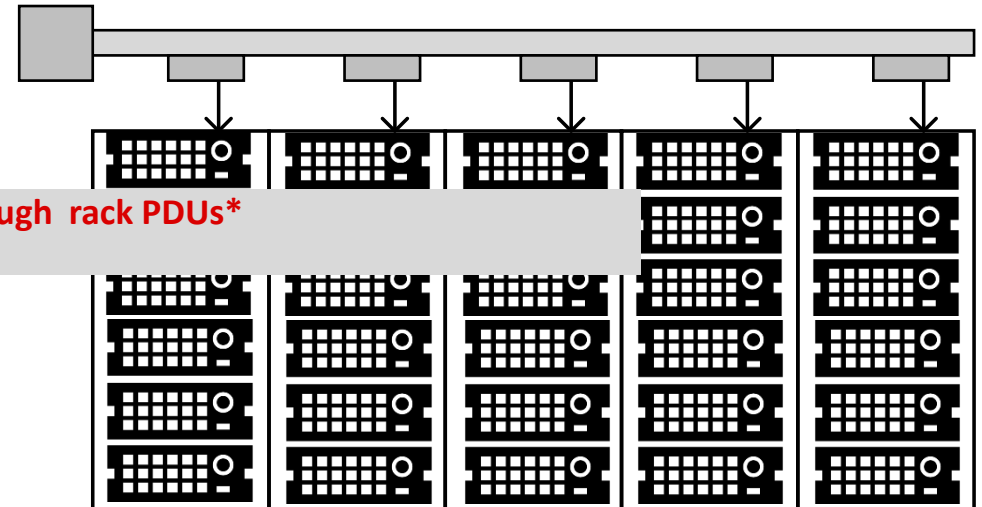
Remote Power Panel (RPP) or Power Distribution Unit (PDU)

- Central low voltage sub-distribution
- Individual cable connections to the server rack
- RPPs supply rack PDUs that are used inside the server cabinets



Busbar+Tapoff units

- Decentralized low voltage sub-distribution
- Electrical distribution via busbars above the server racks
- Each server cabinet has a tap-off unit with a supply line

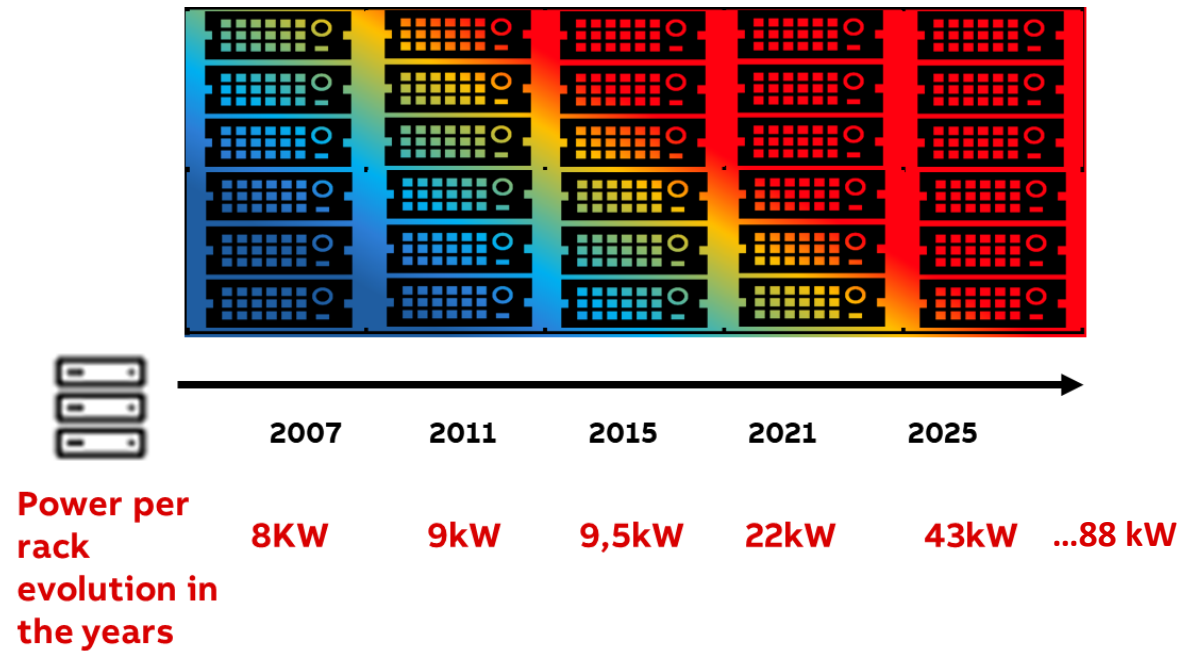


*or power shelves

ABB Rack PDU Solution

Main trends in data centers

Increasing power density



and much more...consider the power needed for AI datacenter

...and increasing protection requirements

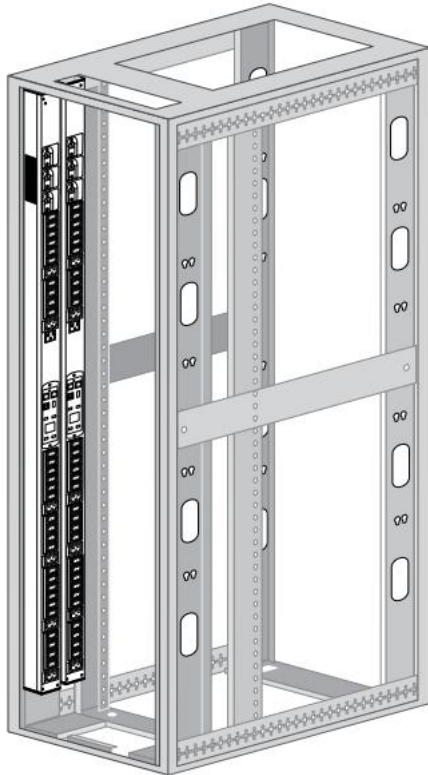
ABB Rack PDU Solution

The facts above mean that...

As power density increases, the rack PDU is becoming the new RPP/ PDC or tap-off unit:
that's why ABB wants to investigate this new opportunity, by offering an essential range of
rack PDU

ABB Rack PDU Solution

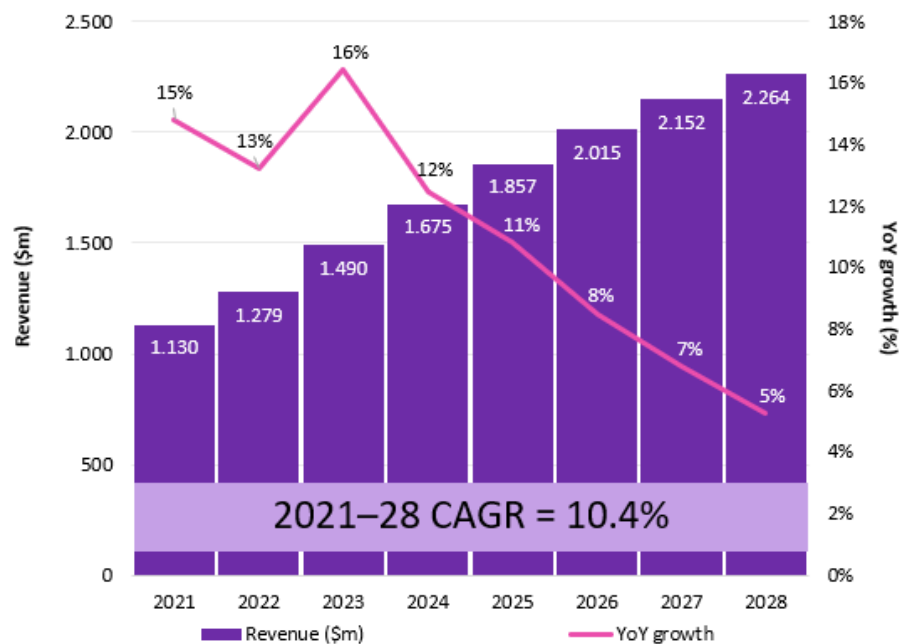
What is a Rack PDU



- In datacenters, a Rack PDU is a device (with multiple outlets) that distributes power to IT equipments in each individual cabinet and rack.
- The purpose of a Rack PDU in networking is to provide a single power source to multiple devices, reducing the need for multiple power sources and simplifying cable management.
- It is typically installed inside the server racks (at the back or side of the rack enclosure)
- The Rack PDU is capable of monitoring, managing, and controlling power consumption to multiple devices in data center environments.

OMDIA Research on rack PDU

Rack PDU market size and growth forecast



Source: Omdia

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Market size and growth of rack PDUs by input phase

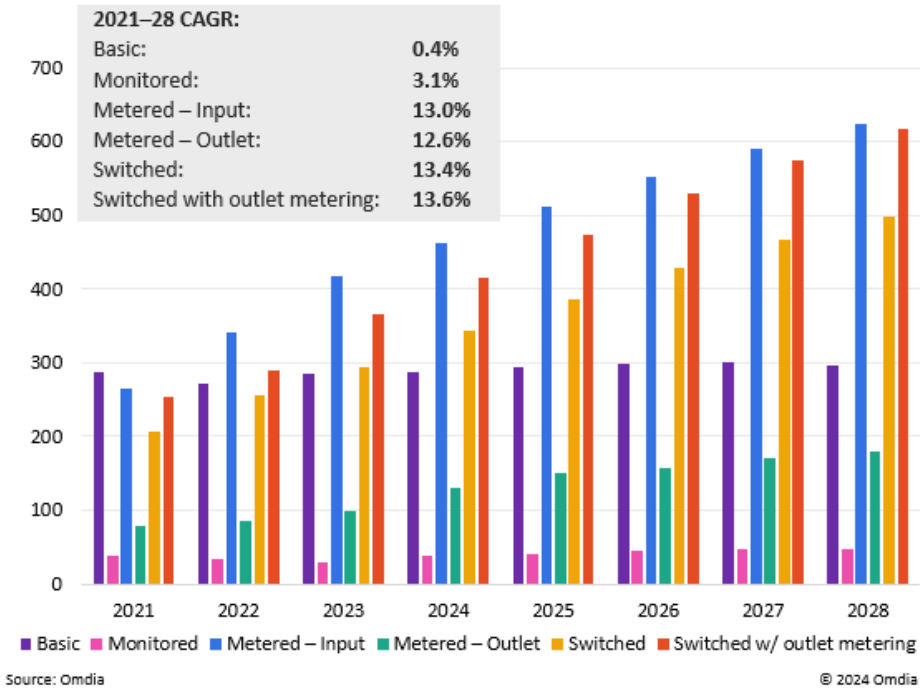


Source: "Data Center Racks Market Analysis 2024 – PDF" - OMDIA research supplied to ABB.
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OMDIA Research on rack PDU

Rack PDU forecast, 2021–28



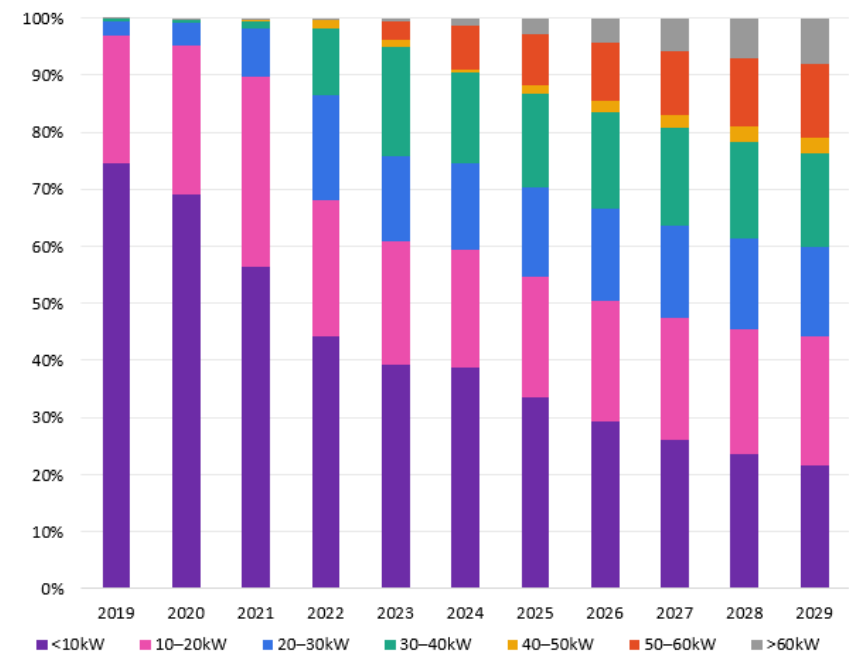
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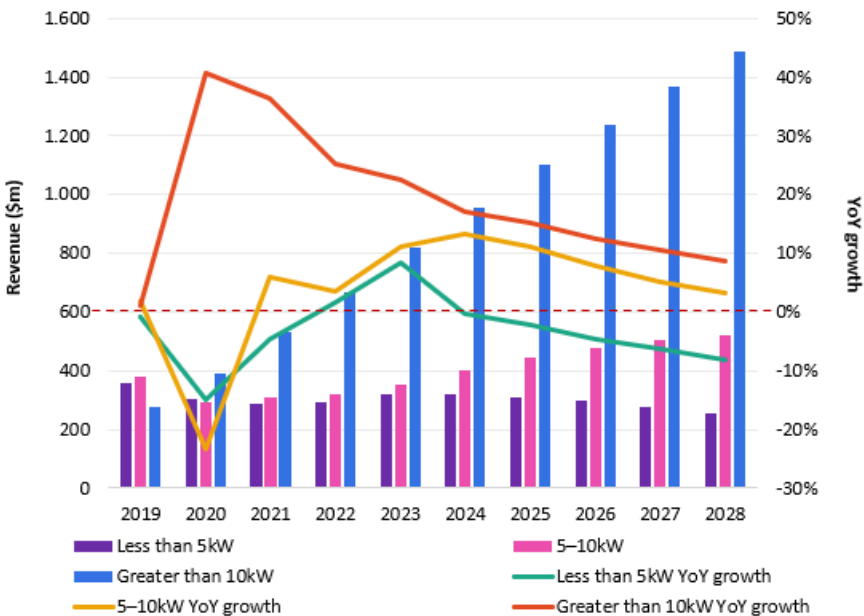


OMDIA Research on rack PDU

Annual DC rack installations split by density



Market size and growth of rack PDUs by power rating



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